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SUPERSEDING
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FEDERAL SPECIFICATION

INVESTMENT, SOLDERING, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. CLASSIFICATION

1.1 **Type and sizes.**—The soldering investment covered by this specification shall be of one type and of the following sizes, as specified in the invitation for bids (see 6.1):

Size 5 (5 pounds)
Size 15 (15 pounds)
Size 50 (50 pounds)
Size 100 (100 pounds)

2. APPLICABLE SPECIFICATIONS

2.1 The following Federal Specification, of the issue in effect on date of invitation for bids, forms a part of this specification:

RR-S-366—Sieves, Standard, Testing.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D. C., or to the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Seattle. Prices may be obtained from the Superintendent of Documents or from the GSA Regional Offices.)

2.2 Specifications and standards applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 **Material.** — The material shall be as hereinafter specified and shall be suitable in every respect to serve the purpose for which intended.

3.2 **Composition.** — Soldering investment shall be a powder, composed essentially of plaster of paris and quartz, or of similar material.

3.3 **Characteristics.**—When the investment is mixed with water in the ratio prescribed by the manufacturer's directions, it shall be satisfactory for the use for which it is intended. The mixed investment shall not crack nor give off poisonous or offensive odors when heated to a minimum of 850° C. (1562° F.) in accordance with 4.3.7, and shall not contaminate the component metal parts embedded in it.

3.4 **Consistency.**—When tested as described in 4.3.1, the average of the minimum and maximum diameters of the slumped mixture shall be not less than 2¼ inches and not more than 3 inches.

3.5 **Setting time.**—The setting time shall be 9 ± 4 minutes, when tested in accordance with 4.3.2.

3.6 **Setting expansion.**—The setting expansion shall be not less than 0.05 percent and not more than 0.30 percent at the end of 24 hours, when tested in accordance with 4.3.3.

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3.7 Thermal expansion. — The thermal expansion shall be not less than 0.50 percent nor more than 1.00 percent when tested in accordance with 4.3.4. The specimen shall not, at any temperature above 200° C. (392° F.), show a length shorter than the original length of the specimen at room temperature.

3.8 Compressive strength. — The compressive strength shall be not less than 900 pounds per square inch, 2 hours after mixing, when tested in accordance with 4.3.5.

3.9 Fineness. — The powder shall be of such fineness that 100 percent will pass the test described in 4.3.6.

3.10 Workmanship. — The workmanship shall be first class in every respect, and the finished product shall be dry, uniform, and free from foreign material and partly set or caked lumps. Coloring material will not be regarded as foreign material.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Unless otherwise specified, at least 3 individual containers of the commodity shall be taken at random from each 100, or fraction thereof, delivered. Samples shall be taken from both the surface and the center of the containers. All laboratory samples shall be placed in an airtight container and sealed.

4.2 Inspection. — Each sample shall be inspected for material, adequacy of directions, workmanship, packaging, packing, and marking.

4.3 Tests.—The investment used in making all test specimens shall be mixed with sufficient distilled water at a temperature of 70° ± 2° F., to conform with the water-powder ratio recommended by the manufacturer. The procedure for mixing shall be as follows: The powder shall be added to the water, and the mixture shall then be mechanically spatulated for 1 minute at the rate of approximately 100 turns of a stiff blade.

4.3.1 *Testing consistency.*—A cylindrical mold, 2 inches long with an internal diameter of 1⅝ inches, shall be placed on a dry glass plate and filled with mixture. Two minutes from the time of starting the mix, the mold shall be lifted and the mixture allowed to slump or spread over the plate. Approximately 1 minute after the cylinder is lifted, the smallest and largest diameter of the slumped mixture shall be determined.

4.3.2 *Setting time.*—The time of set shall be determined with a Vicat apparatus consisting of a frame bearing a movable rod weighing 300 grams. One end of the rod shall be 1 cm. in diameter for a distance of at least 5 cm., and the other end shall have a removable needle 1 mm. in diameter and 5 cm. in length. The rod shall be reversible and shall have an adjustable indicator which moves over a scale (graduated in mm.) attached to the frame. The rod shall be held in any desired position by a screw attached to the frame. The paste is held in a rigid conical ring resting on a glass plate about 10 cm. square. The ring shall be made of a noncorroding, nonabsorbent material, having an inside diameter of 7 cm. at the base, 6 cm. at the top, and a height of 4 cm. In addition the Vicat apparatus shall conform to the following requirements:

Weight of plunger 300 gm. ± 0.5 gm.
Diameter of needle 1 mm. ± 0.05 mm.

A 300-gram sample, mixed as specified in 4.3, shall be placed in the conical ring and tested with the 300-gram Vicat needle. The needle shall be allowed to sink into the paste at 2-minute intervals until nearly set, and at 1 minute intervals thereafter. After each penetration, the needle shall be wiped clean, and the mold moved slightly so that the needle will not strike the same place twice. Setting time is considered complete when the needle no longer penetrates to the bottom of the specimen.

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4.3.3 *Setting expansion.* — Prepare the specimen and determine the length. Let the specimen stand for 24 hours, measure again, and calculate the change in length based on the length made immediately after the setting time. The change in length on setting shall be determined on a specimen approximately 30 centimeters long by observing the change with a micrometer microscope comparator, or equipment of comparable accuracy.

4.3.4 *Thermal expansion.* — Thermal expansion shall be measured on a specimen approximately 1.2 centimeters in diameter by approximately 20 centimeters in length, by the "fused quartz expansion apparatus" method, or by the use of equipment of comparable accuracy. The specimen shall be measured approximately 2 hours after the beginning of the mix. Subsequent measurements shall be made at 10-minute intervals. Heating shall start 2 hours after beginning the mix. The specimen shall be gradually heated from room temperature 21.1° ± 1.11° C. (70° ± 2° F.) to 700° C. (1292° F.) in 3 hours. The thermal expansion computed shall be the increase in length of the specimen from beginning of heating to end of heating.

4.3.5 *Compressive strength.*—One set of specimens shall be prepared and tested 2 hours after mixing. The compressive strength shall be determined on five cylindrical specimens not less than 1 3/10 inches in diameter and not less than 2 inches in length. The vessel containing the mix shall be moved back and forth over the mold while pouring. The specimens shall be worked slightly to remove air bubbles and then be leveled off. The cylinders shall be removed from the molds as soon as they are hard enough to handle, and stored in air at a temperature of not less than 15° C. (59° F.) and not more than 38° C. (100° F.).

4.3.5.1 The compressive strength of the samples shall be the arithmetic average of the recorded values for the 5 cylinders tested or the recomputed arithmetic average if discarding is necessary. In computing the compressive strength, any cylinder having a recorded value which varies more than 15 percent from the computed average shall be discarded and the average of the remaining samples shall be recomputed. In case three or more cylinders vary from the computed average more than 15 percent, the samples shall be discarded and the test repeated. Failure of the second set of samples constitutes failure of the lot represented by the samples.

4.3.6 *Fineness.*—The sample of investment shall be spread out in a thin layer and dried at 45° C. (113° F.) for 2 hours and then cooled in a moisture-free atmosphere. Fineness shall be determined by screening not less than 100 grams (3½ ounces) of the dried sample through a standard No. 30 sieve conforming to the requirements of Federal Specification RR–S–366. The material shall be shaken for 5 minutes on a Ro-tap or equivalent machine at a speed of 290 revolutions per minute and 156 taps per minute.

4.3.7 *Heat test.* — Five specimens of approximately 0.5 by 1.5 by 2.0 inches each shall be prepared for subjection to heat testing. After pouring the specimens they shall be stored in air at room temperature for 1 hour. Immediately thereafter the specimens shall be subjected to the following heat cycle:

(*a*) Place in furnace at 427° ± 10° C. (800° F.) and raise the furnace temperature to 850° ± 10° C. in 45 minutes.

(*b*) Hold at 850° C. for 5 minutes.

(*c*) Remove specimen to room temperature for 1 minute.

(*d*) Replace in furnace at 850° C. for 5 minutes.

(*e*) Remove specimen from furnace and immediately examine visually for cracks.

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If three or more of the five specimens show cracks, the material shall be reported as failing to meet this requirement.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Unless otherwise specified, soldering investment shall be packaged in accordance with accepted commercial practice. The container in which the soldering investment is packaged shall be such that it will not be adversely affected thereby.

5.2 Packing.—Soldering investment shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate, and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Packages.*—Unless otherwise specified, each package shall be marked with the name and size of the commodity, quantity contained therein, name or registered trade-mark of the manufacturer and contractor, water-powder ratio, setting expansion, crushing strength, setting time, thermal expansion at approximately 700° C., and directions for mixing and use.

5.3.2 *Shipping containers.*—Unless otherwise specified, each shipping container shall be marked with the name of the commodity as defined by the contract or order under which shipment is made, the quantity contained therein, the name of the contractor, the number of the contract or order, the name and address of the consignee, and the gross weight.

6. NOTES

6.1 Intended use.—Soldering investment is intended for use in dental clinics and dental laboratories for forming accurate dental prosthetic models, and for investing the component parts of bridges and the components of other dental prosthetic appliances to hold them in proper relationship for soldering.

6.2 Ordering data. — Purchasers should specify the size required and should exercise any desired options offered herein. (See 1.1, 4.1, 5.1, 5.3, 6.3, and 7.2.6.)

6.3 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically requested in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.4 This specification covers the following stock numbers listed in the Armed Services Catalog of Medical Materiel:

5–368–050—Investment, Soldering, 5-Pound.

5–368–550—Investment, Soldering, 15-Pound.

6.5 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, sizes, etc., normally procured for Federal Government requirements.

6.6 Transportation description.

6.6.1 *Transportation description applicable to these items is:*

Dental investment compound.

*In bulk in 5-ply paper bags, in barrels or steel pails.

Carload weight, 40,000 pounds

Truckload weight, 40,000 pounds

* Preferred method of shipment to obtain lowest transportation charge.

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In containers in barrels or boxes, or in metal cans in wirebound crates.
Carload weight, 36,000 pounds
Truckload weight, 36,000 pounds

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications and standard, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments:

7.2 Army, Navy, and Air Force.

7.2.1 *Applicable specifications and standard.*

Federal Specifications:

NN-B-601—Boxes; Wood-Cleated-Plywood, for Domestic Shipment.

NN-B-621—Boxes; Wood, Nailed and Lock-Corner.

NN-B-631—Boxes; Wood, Wirebound (for Domestic Shipment).

LLL-B-631—Boxes; Fiber, Corrugated (for Domestic Shipment).

Military Specifications:

JAN-P-105—Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood.

JAN-P-106—Packaging and Packing for Overseas Shipment — Boxes; Wood, Nailed.

MIL-B-107—Boxes, Wood, Wirebound (Overseas Type).

JAN-P-125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.

MIL-A-140—Adhesive, Water-Resistant, Waterproof Barrier-Material.

Navy Department Specification:

General Specifications for Inspection of Material.[1]

Military Standard:

MIL-STD-129—Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.2 *Size 5 (5-pound container).*

7.2.2.1 *Packaging.*

7.2.2.1.1 *Unit packages.*—Five pounds of soldering investment constitutes one unit. Each unit shall be packaged in a metal container of suitable size and design, fitted with a double friction-type closure. The container in which the soldering investment is packaged shall be such that it will not adversely affect or be adversely affected by the container contents.

7.2.2.1.2 *Intermediate packages.*—Six containers (6 units) shall be packaged in a carton of appropriate size conforming to the requirements of Federal Specification LLL-B-631. Carton design shall include corrugated fiberboard liner, top and bottom pads, and full-length partitions. As an alternate to the above, each unit may be individually packaged in individual corrugated fiberboard cartons conforming to the requirements of Federal Specification LLL-B-631.

[1] Applicable to Navy purchases only.

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7.2.2.2 *Packing.*

7.2.2.2.1 *Domestic.* — Two intermediate packages (12 units) (or 12 containers if packaged in individual cartons) shall be packed in a shipping container conforming to the requirements of Federal Specification NN-B-601, NN-B-621, or NN-B-631.

7.2.2.2.2 *Overseas.* — Two intermediate packages (12 units) (or 12 containers if packaged in individual cartons) shall be packed in a shipping container designed for type 2 load and conforming to the requirements of Military Specification JAN-P-105, JAN-P-106, or MIL-B-107. Closure and strapping shall conform to the requirements of the appendix to the applicable box specification. Each shipping container shall be lined with a sealed waterproof bag, or its equivalent, made from material conforming to the requirements of Military Specification JAN-P-125, for case liners; seams and closures shall be sealed with a continuous seam at least ¾ inch wide of water-resistant adhesive conforming to the requirements of Military Specification MIL-A-140.

7.2.2.2.3 *Packing variation.*—If the number of units in any shipment is not the exact multiple of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in a shipping container of suitable size and design, and so constructed as to insure safe delivery to the specified destination.

7.2.3 *Size 15 (15-pound container).*

7.2.3.1 *Packaging.*

7.2.3.1.1 *Unit packages.*—Fifteen pounds of soldering investment constitutes one unit. Each unit shall be packaged in a metal container of suitable size and design, fitted with a double friction-type closure. The container in which the soldering investment is packaged shall be such that it will not adversely affect or be adversely affected by the container contents.

7.2.3.1.2 *Intermediate packages.* — Each container (1 unit) shall be packaged in a corrugated fiberboard carton of suitable size and design, conforming to the requirements of Federal Specification LLL-B-631. Carton design shall include corrugated fiberboard liner, top and bottom pads, and full-length partitions.

7.2.3.2 *Packing.*

7.2.3.2.1 *Domestic.* — Four intermediate packages (4 units) shall be packed in a shipping container conforming to the requirements of Federal Specification NN-B-601, NN-B-621, or NN-B-631.

7.2.3.2.2 *Overseas.* — Four intermediate packages (4 units) shall be packed in a shipping container designed for type 2 load and conforming to the requirements of Military Specification JAN-P-105, JAN-P-106, or MIL-B-107. Closure and strapping shall conform to the requirements of the appendix to the applicable box specification. Each shipping container shall be lined with a sealed waterproof bag, or its equivalent, made from material conforming to the requirements of Military Specification JAN-P-125, for case liners; seams and closures shall be sealed with a continuous seam at least ¾ inch wide of water-resistant adhesive conforming to the requirements of Military Specification MIL-A-140.

7.2.3.2.3 *Packing variation.*—If the number of units in any shipment is not the exact multiple of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in a shipping container of suitable size and design, and so constructed as to insure safe delivery to the specified destination.

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7.2.4 *Marking.*

7.2.4.1 *Packages.*—Each unit package and each intermediate package shall be marked with the stock number, nomenclature, unit, quantity, and name of the contractor. When the contractor is not the manufacturer, the name of the manufacturer shall also appear as the manufacturer. In addition, each package shall contain the marking information specified in 5.3.1.

7.2.4.2 *Shipping containers.*—Each shipping container shall be marked in accordance with Military Standard MIL–STD–129 and ASMPA Marking Instructions.

7.2.5 *Inspection procedures (Navy only).*—The general inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

7.2.6 *Ordering data.*—Procurement documents should specify whether the subject commodity is to be packed for domestic and/or overseas shipment. (See 7.2.2.2 and 7.2.3.2).

NAVY INTEREST: Md

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

For sale by the Superintendent of Documents, Washington 25, D. C. Price 5 cents.

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